

OSCAR R MOLL

AI Research Engineer

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SUMMARY

AI research engineer with an MIT PhD in computer science focused on systems and ML methods for image and video search, with first-author papers at SIGMOD and ICDE. Experienced in evaluation-driven research spanning benchmark and dataset design, error analysis, model and algorithm development, and distributed experimentation infrastructure. Production engineering experience on Databricks Knowledge Assistant, an enterprise RAG question-answering agent.

EXPERIENCE

Databricks

Software Engineer (full time)

Dec 2024–Jun 2026

Knowledge Assistant

- Reduced p99 time to first token (TTFT) from minutes to seconds by replacing per-assistant single-tenant serving with a shared, multi-tenant control-plane service, eliminating pod cold starts. Refactored Python state and credential propagation for tenant-safe execution and supported both architectures during a region-by-region migration.
- Increased Knowledge Assistant’s release cadence from manual releases every few weeks to automated daily staging and twice-weekly production deployments by extracting its CRUD APIs from a shared multi-product service. Migrated roughly 15 dependency configurations to a new Java RPC framework, added real-dependency deployment gates, and moved all production traffic.
- Delivered a highly requested feature enabling customers to add, index, update, and remove data sources after agent creation without recreating the agent, re-ingesting unchanged documents, or changing its endpoint; aligned the user flow and APIs with product, UX, and frontend partners.
- Drove Knowledge Assistant adoption by a strategic enterprise customer as the sole Knowledge Assistant engineer on the engagement, coordinating with Field Engineering and the PDF parsing team to meet document-indexing scale and accuracy requirements.
- Reduced weekly document-ingestion failure triage from hours to minutes by conceiving and building a hackathon tool with two coworkers; it gathered and summarized logs, grouped failures, and updated Jira tickets. Drove adoption by processing a 100+ ticket backlog.

vlm.run

Applied AI Engineer (part-time)

Aug 2024–Nov 2024

- Raised VisualWebArena benchmark scores for an image-based browser agent built on Playwright within a three-person team by triaging failed trajectories, inspecting fully assembled (“compiled”) model prompts, fixing benchmark and action-parsing errors, and experimenting with multimodal browser-state representations and concise state histories.
- Built a notebook-based evaluator for schema-constrained image-to-JSON extraction, creating an initial set of manually labeled examples and basic type-aware scoring rules for numeric, categorical, and nested fields.

Pixeltable

Aug 2023–May 2024

- Enabled full-system demo notebooks to run on either Google Colab or a laptop, rather than requiring a separately provisioned PostgreSQL server. To achieve this, built cross-platform PostgreSQL and pgvector binary wheels plus management routines to initialize and run PostgreSQL across operating systems and root or non-root environments.
- Built initial AI ecosystem integrations to bring data in and out of Pixeltable: imported Hugging Face and Parquet datasets, extracted and chunked PDF text, and exported transformed and filtered Pixeltable query results as sharded PyTorch `IterableDatasets` ready for multi-worker training with low I/O overhead.

MIT CSAIL, Data Systems Group

Postdoctoral Research Assistant

Jul 2023–Mar 2024

Cambridge, MA

- Built an evaluation-driven quality loop for an LLM pipeline that extracted mathematical-model information from scientific papers: assembled development examples, traced generated results back to source-PDF regions, and used failure analysis to guide prompt and parser improvements.
- Reduced the pipeline’s iteration latency and failure surface by parallelizing LLM requests, replacing remote ontology lookups with local cached retrieval, enforcing output schemas, and adding end-to-end integration tests.
- Mentored a master’s student developing AI methods to extract numerical data from scientific plots, helping create labeled development and evaluation sets in Roboflow to make quality measurable and guide iteration.

- Conceived and built ExSample, an adaptive system for cold-start object search over unindexed video that reduced redundant near-duplicate results, by combining object detectors, random selection and a novel estimator of a video's expected yield of unseen physical instances. Across 5 datasets and 34 object queries, ExSample used up to 6x fewer calls to expensive object detectors than a strong sampling+skipping baseline, and for 10% limit queries, it returned results between 30x and 10000x sooner than waiting for fast CNN-based models to score and rank the repository.
- Reformulated the system evaluation after finding that metrics such as frame-level recall were dominated by repeated views of long-visible instances. Instead, designed the system to optimize the number of unique instances found for a given number of object-detector calls.
- Built the required instance-labeled video datasets despite scarce ground truth. Iterated on object-track quality by fine-tuning object detectors in PyTorch on a small set of manually labeled examples from each target dataset. Validated the generated object-tracks through visual inspection and comparison of object-duration histograms for model-generated pseudo-labels with those for manually labeled videos.
- Conceived and built SeeSaw, an interactive image-search system that helps users recover when zero-shot search with pretrained vision-language embeddings falls short. In a 40-person study, SeeSaw reduced median time to find 10 relevant results to about 2 minutes, versus more than 6 minutes with the zero-shot baseline.
- Drove the evaluation-backed development loop from problem discovery through system improvement: identified a long tail of failures hidden by impressive search demos, built a benchmark spanning 1,000+ queries across four datasets, and translated the resulting error analysis into a two-pronged approach combining a custom relevance-feedback algorithm – formulated as a regularized optimization problem and solved in PyTorch at each feedback iteration – with an enriched multiscale image representation. The combined system improved mean average precision of search results from 0.19 to 0.46.
- Built the research platform to support a fast development loop: a Python+Ray-based distributed benchmark runner for 1,000+ searches to evaluate algorithmic changes, a distributed Ray+PyTorch pipeline for experimenting with different image representations and embeddings, and a serving system and UI for a 40-person study to measure gains when used by real people.

SELECTED EARLIER EXPERIENCE

Tesla Autopilot

Machine Learning Engineer (PhD Intern)

Jun 2017–Aug 2017

Palo Alto, CA

- Designed and implemented a fleet-scale telemetry monitoring tool to flag potential map gaps through statistical tests of distribution change. Deployed it as a Spark streaming pipeline and traced worker OOMs to geographic skew during group aggregation. Resolved the OOMs with a custom Spark Aggregator that bounded partial sample sizes before the shuffle and merged them afterward while preserving the distribution of global uniform sampling without replacement.

Google

Software Engineer (PhD Intern)

Jun 2016–Aug 2016

Mountain View, CA

- Investigated reducing deserialization overhead in microservice architectures by preventing full message deserialization when possible. Identified CPU work and memory allocation as deserialization bottlenecks and addressed them with a custom expression evaluator that worked directly on the byte stream, extracting only needed fields. Implemented an LLVM-generated evaluator that achieved 4.2–9x the throughput of traditional C++ Protobuf deserialization in expression-evaluation microbenchmarks.

Amazon Web Services

Software Development Engineer (full time)

Sep 2012–Jan 2015

Seattle, WA

- Designed and implemented a decentralized, asynchronous Write-Ahead Log (WAL) replication system to bring Amazon Aurora WAL server replicas up to date after failures. Keeping all replicas up to date improves system resilience and throughput after failures. Owned the inter-replica RPC APIs and their integration into an evolving multithreaded, asynchronous C++ system.

EDUCATION

Massachusetts Institute of Technology

PhD, Computer Science '23

MEng, Computer Science '12. SB, Computer Science and Mathematics '11

Cambridge, MA

SELECTED PUBLICATIONS

- Oscar Moll et al. “SeeSaw: Interactive Ad-hoc Search Over Image Databases.” *Proceedings of the ACM on Management of Data, SIGMOD*, 2023.
- Oscar Moll et al. “ExSample: Efficient Searches on Video Repositories through Adaptive Sampling.” *IEEE International Conference on Data Engineering (ICDE)*, 2022.
- Coauthor of additional work on GPU-oriented database execution, hardware lock elision, and multimodal data systems in PVLDB, DISC, and related venues.